MILITARY SPECIFICATION

CLOTHING, CHEMICAL AGENT PROTECTIVE,

5 AND 10 MAN OUTFITS; TREATMENT, PACKAGING AND PACKING OF

1. SCOPE

1.1 Scope.- This specification covers the requirements for the treatment by impregnation, of chemical agent protective clothing, 5 and 10 man outfits, and their packaging, packing and marking for shipment and storage.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-S-48	- Socks, Men's, Wool, Cushion Sole, Stretch Type
MIL-P-265	- Polyvinyl Alcohol, Granular
MIL-I-285	- Impregnite, Unstabilized (CC2) and Stabilized (XXCC3)
MIL-C-429	- Chlorinated Paraffin, Technical
MIL-G-1057	- Gloves, Cloth, Cotton, Special, OG-109
MIL-B-17757	- Boxes, Fiberboard, Corrugated (Modular Size)
MIL-H-29115	- Hood, Chemical Protective
MIL-L-43579	- Liner, Trousers, Chemical Protective
MIL-L-43582	- Liner, Shirt, Chemical Protective
MIL-B-43587	- Bag, Chemical Protective Clothing

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	- Marking for Shipment and Storage

FSC 8415

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PUBLICATIONS

EDGEWOOD ARSENAL TECHNICAL MEMORANDUM

EATM 311-31 - Laboratory Methods for Evaluating the Protection of Permeable Protective Clothing Against Chemical Agents

(Rocky Mountain Arsenal Operating Procedure No. SMURM-O-P-8).

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issues in effect on date of invitation for bids or request for proposal shall apply:

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

NATIONAL MOTOR FREIGHT CLASSIFICATION

(Application for copies should be addressed to the American Trucking Associations, Attn: Tariff Order Section, 1616 P Street, N.W., Washington, D.C. 20036.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

UNIFORM FREIGHT CLASSIFICATION

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, Illinois 60606.)

3. REQUIREMENTS

3.1 Component articles.- Each chemical protective clothing liner outfit shall consist of one impregnated shirt, one pair of impregnated trousers, one impregnated hood, one pair of impregnated gloves and one pair of impregnated socks, contained in a special closure type polyethylene bag imprinted with use instructions. The individual size of component articles to be included with each size of outfit shall be as specified in table I.

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Table I - Size of component article

Liner outfit size	Component article sizes				
	Shirt	Trousers	Gloves	Socks	Hood
Small	Small	Small	Large	Large	One size
Medium	Medium	Medium	Large	Large	One size
Large	Large	Large	Large	Large	One size

3.2 Impregnation loading.- The clothing components of the chemical protective clothing liner outfit shall conform to the impregnation loading characteristics of table II when tested as specified in 4.4.4.1.1, 4.4.4.2.1 and 4.4.4.2.2.

Table II - Impregnation loading

Component article	Active chlorine content (mg/sq cm)	Chlorinated paraffin content (% by wt.)	Weight pickup (% by wt.)
Shirt	0.6 ± 0.1	30 ± 5	-
Trousers	0.6 ± 0.1	30 ± 5	-
Gloves	1.1 ± 0.3	30 ± 10	-
Socks	1.1 ± 0.3	-	48 ± 16
Hood	0.6 ± 0.1	30 ± 5	-

3.3 Materials.-

3.3.1 Chemicals.- The chemicals used in the impregnating suspension shall conform to table III. No other chemicals shall be used.

Table III - Chemicals

Chemicals	Conform to
Impregnite (XXCC3)	Type II, MIL-I-285
Chlorinated paraffin	Type I, MIL-C-429
Polyvinyl alcohol	Grade A, Class 1, MIL-P-265

3.3.2 Liner outfit articles.- The unimpregnated clothing shall conform to table IV.

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Table IV - Liner outfit articles

Liner outfit articles	Conform to
Shirt	MIL-L-43582
Trousers	MIL-L-43579
Gloves	MIL-C-1057
Socks	MIL-S-48
Hood	MIL-H-29115

3.4 Post washout loss.- After each batch (see 6.3) of dried, impregnated hoods, shirts or trousers is post washed for 15 minutes in clear water at 90°F and then dried, the loss of impregnation chemicals shall not be more than 15 percent of impregnated clothing when tested as specified in 4.4.4.2.3.

3.5 Mustard penetration. - Mustard penetration of impregnated clothing shall not be more than 2 micrograms per square centimeter when tested as specified in 4.3.4.1.

3.6 Workmanship.- The impregnated clothing shall be free from excessive build-up of impregnite (white area), surface harshness, grease, dirt and other foreign matter.

3.7 First article.- Prior to the start of regular production, a separate first article sample of impregnated liner outfits shall be produced in accordance with this specification for examination and test (see 4.3 and 6.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Objective evidence.- The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.3 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Classification of inspection.- The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

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4.3 First article inspection

4.3.1 Lot.- A first article inspection lot of impregnated liner outfits consisting of a number of liner outfits as specified in contract or purchase order (see 6.2) shall be produced, using the same methods, materials, equipment and processes as will be used during regular production.

4.3.2 Sampling.-

4.3.2.1 For examination.- Sampling of impregnated liner outfits shall be conducted in accordance with MIL-STD-105.

4.3.2.2 For nondestructive tests.- The sample for nondestructive tests shall consist of each entire batch of impregnated articles.

4.3.2.3 For destructive tests.-

(a) Active chlorine content.- Sampling of each batch of impregnated clothing shall be conducted in accordance with table V. The sample dimensions shall be 38.7 square centimeters or six (6) square inches.

(b) Mustard penetration.- Sampling of each batch of impregnated hoods, shirts and trousers shall be conducted in accordance with table V. The sample test shall be 100 square centimeters.

4.3.3 Inspection procedure.-

4.3.3.1 For examination.- The sample of impregnated liner outfits shall be examined in accordance with the classification of defects specified in 4.4.3.4 and with MIL-STD-105.

4.3.3.2 For nondestructive tests.- Samples taken as specified in 4.3.2.2 shall be tested in accordance with 4.4.4.2.1, 4.4.4.2.2 and 4.4.4.2.3. The acceptance number is zero.

4.3.3.3 For destructive tests.- The sample of each batch of impregnated article of clothing shall be tested in accordance with 4.3.4, 4.4.4.1 and 4.4.4.1.1. The acceptance number is zero.

4.3.4 Test.-

4.3.4.1 Mustard penetration.- Mustard penetration test shall be conducted in accordance with EATM 311-31, section III, paragraph A except the time of test shall be 2 hours.

4.4 Quality conformance inspection.-

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4.4.1 Lotting.- A lot shall consist of the impregnated outfits from acceptable batches of each component article (see 4.4.2.2) produced during one week, by one manufacturer, at one plant, from the same materials, under essentially the same manufacturing conditions.

4.4.2 Sampling.-

4.4.2.1 For examination.- Sampling of impregnated liner outfits shall be conducted in accordance with MIL-STD-105.

4.4.2.2 For nondestructive tests.- The sample for nondestructive tests shall consist of each entire batch of impregnated articles.

4.4.2.3 For destructive test.-

4.4.2.3.1 Active chlorine content.- Sampling of each batch of impregnated clothing shall be conducted in accordance with table V. The sample dimensions shall be 38.7 square centimeters or six (6) square inches.

Table V - Sampling for destructive tests

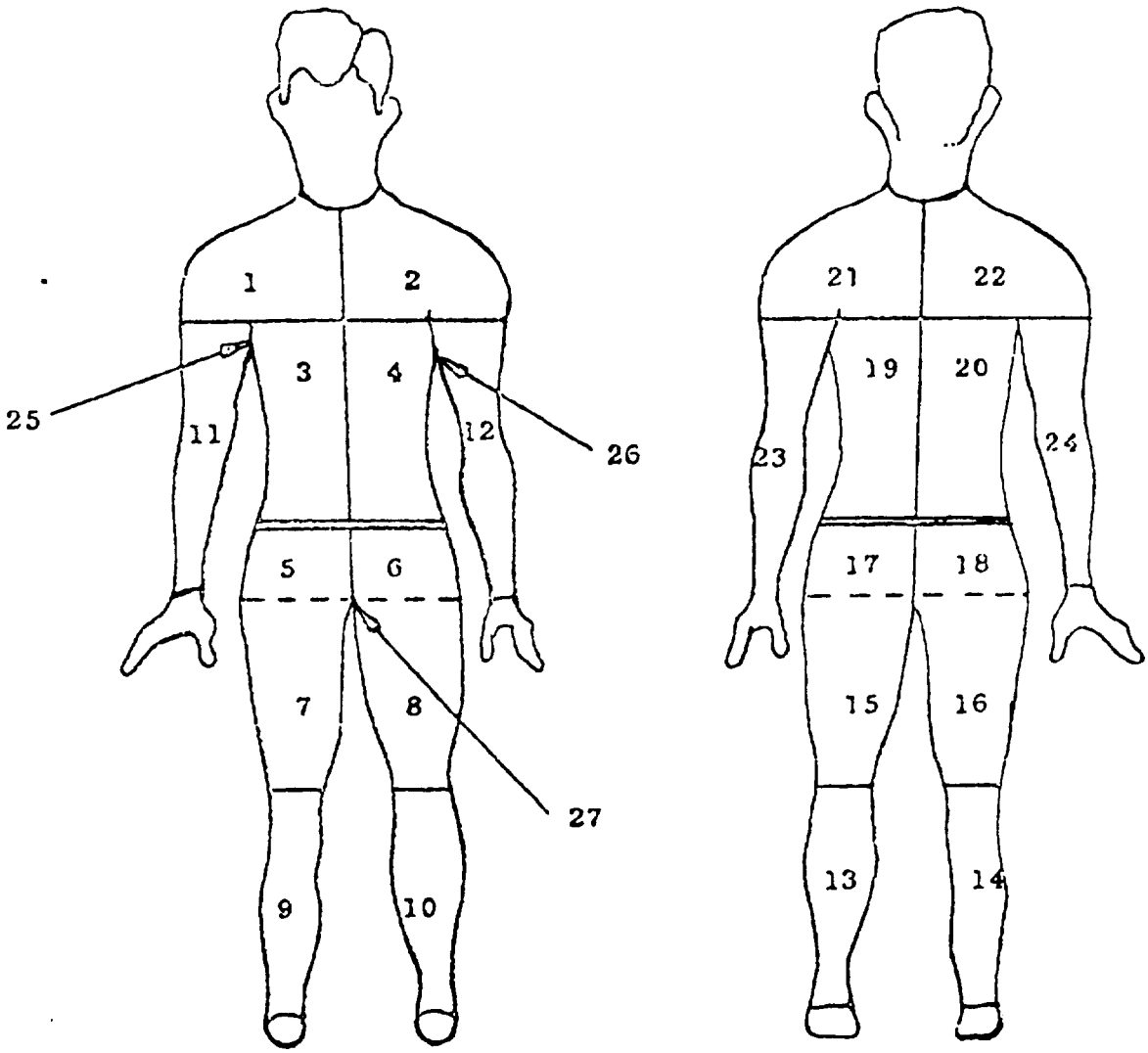
Component article	Number of articles	Sampling positions for each article: Position numbers in figure 1	Position	Total number of samples
Shirt	2	21, 22, 23, 24	---	8
Trousers	2 pairs	7, 8, 17, 18	---	8
Gloves	2 pairs	---	Front and back of palm from each glove	8
Socks	2 pairs	--	Two samples from each sock taken below the ribbed top and above the heel	8
Hood	2	---	Two samples from each hood taken from front skirt area and top back hood area	8

4.4.3 Inspection procedure.-

4.4.3.1 For examination.- The sample impregnated liner outfits shall be examined in accordance with the classification of defects as specified in 4.4.3.4 and with MIL-STD-105.

4.4.3.2 For nondestructive tests.- Samples taken as specified in 4.4.2.2 shall be tested in accordance with 4.4.4.2.1, 4.4.4.2.2 and 4.4.4.2.3. The acceptance number is zero.

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CHEMICAL PROTECTIVE CLOTHING LINER OUTFIT
SAMPLING POSITIONS

FIGURE 1

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4.4.3.3 For destructive tests.- The sample of each batch of impregnated articles of clothing shall be tested in accordance with 4.4.4.1. The acceptance number is zero.

4.4.3.4 Classification of defects

(a) Clothing liner outfit, chemical protective

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Components and sizes incorrect (3.1)
102	Workmanship (3.6)

(b) Preparation for delivery (see Section 5)

Categories	Defects
Critical:	None defined
Major:	AQL 2.5 percent defective
101	Incomplete heat sealing of bag
102	Excess air in sealed bag
104	Label incorrect
105	Tears or holes in bag
106	Containers not as specified
107	Strapping incorrect

4.4.4 Tests.-

4.4.4.1 Destructive tests.- The samples specified in table V shall be cut from each component of impregnated article. The sample shall not include seams, buttonholes or any area of more than one fabric thickness. Confined areas such as inner parts of arm or crotch seams shall be avoided. Tests shall then be conducted as specified in 4.4.4.1.1. Distilled water and reagent grade chemicals shall be used throughout the test.

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4.4.4.1.1 Active chlorine content.- Active chlorine content shall be determined as follows:

(a) Calculate the area of the sample to be tested in square centimeters.
(b) Cut this sample into pieces approximately 1/4 inch by 1 inch.
(c) Put these pieces into a 300-milliliter (ml) glass-stoppered Erlenmyer flask.
(d) Add 25 ml of chloroform.
(e) Add 50 ml of 5 percent potassium iodide solution.
(f) Add 10 ml of the ratio one to one solution of hydrochloric acid.
(g) Titrate this with 0.1 normal solution of sodium thiosulfate until solution has a light yellow color.
(i) Add 5 ml of starch indicator solution.
(j) Continue the titration until the solution remains colorless.
(k) Calculate the amount of active chlorine by the following formula:

$$A = \frac{B \times C \times 17.7}{D}$$

where: A = Milligrams active chlorine per square centimeter,
B = Sodium thiosulfate solution used for the titration in ml,
C = Normality of sodium thiosulfate solution, and
D = Area of the sample in square centimeters.

(l) Determine the active chlorine content of the batch by averaging the results of the eight samples.

4.4.4.2 Nondestructive tests.-

4.4.4.2.1 Weight pick-up.- Weight pick-up shall be determined as follows:

(a) Weigh the total dry batch to be impregnated.
(b) Weigh the entire batch after it has been impregnated and dried.
(c) Calculate the percentage of weight pick-up by the following formula:

$$C = \frac{B-A}{A} \times 100$$

where: A = Weight of the batch to be impregnated,
B = Weight of the batch after it has been impregnated and dried, and
C = Percentage of weight pick-up.

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4.4.4.2.2 Chlorinated paraffin content.- The chlorinated paraffin content in the impregnated clothing shall be determined as follows:

(a) Weigh the total dry batch to be impregnated.
(b) Weigh the entire batch after it has been impregnated and dried.
(c) Calculate the percentage of the chlorinated paraffin content in the impregnated clothing by the following formula:

$$D = \frac{(B-A)\ C}{A}$$

where A = Weight of the total dry batch to be impregnated.
B = Weight of the entire batch after it has been impregnated and dried,
C = Percentage of chlorinated paraffin content by weight of the solids in the suspension, and
D = Percentage of chlorinated paraffin content in the impregnated clothing.

4.4.4.2.3 Post washout loss.- Post washout loss for the impregnated hoods, shirts or trousers shall be determine as follows:

(a) Weigh the entire batch of hoods, shirts or trousers to be impregnated,
(b) Weigh the impregnated and dried batch,
(c) Weigh the post washed and dried batch,
(d) Calculate the percentage of post washout loss by the following formula:

$$D = \frac{B-C}{A} \times 100$$

where A = Weight of the entire batch of hoods, shirts or trousers to be impregnated,
B = Weight of the impregnated and dried batch,
C = Weight of the post washed and dried batch, and
D = Percentage of post washout loss.

5. PREPARATION FOR DELIVERY

5.1 Packaging.- Packaging shall be level A (see 6.2).

5.1.1 Level A.- Each complete XXCC3 impregnated outift, consisting of one shirt liner, trouser liner, hood and one pair each of socks and gloves (see 3.1) shall be packaged in a special closure type polyethylene bag conforming to type III of MIL-B-43587.

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5.2 Packing (for 5-man outfit).- Packing shall be level A or B as specified (see 6.2).

5.2.1 Level A.- Five (5) outfits (see table VI), unit packaged as specified in 5.1, shall be packed in a fiberboard shipping container assembled, closed and reinforced conforming to class weather-resistant, size 2A of MIL-B-17757. The fiberboard for the box liner shall conform to type CF, class weather-resistant, variety DW, grade V15c of MIL-B-17757. Level A bundles shall be packed flat, two in length, one in width and two in depth with the fifth bundle placed in the center of the container. The length of the bundles shall be parallel to the end panels of the container. The voids within the container shall be filled with cushioning material to immobilize the outfits.

5.2.2 Level B.- Five (5) outfits (see table VI) unit packaged as specified in 5.1, shall be packed in a fiberboard shipping container, assembled, closed and reinforced conforming to class domestic, grade 275, size 2A of MIL-B-17757. The fiberboard for the box liner shall conform to type CF, class domestic, variety DW, grade 200 of MIL-B-17757. Level A bundles shall be packed flat, two in length, one in width and two in depth with the fifth bundle placed in the center of the container. The length of the bundles shall be parallel to the end panels of the container. The voids within the container shall be filled with cushioning material to immobilize the outfits.

5.3 Packing.- Packing (for 10-man outfit).- Packing shall be level A or B as specified (see 6.2).

5.3.1 Level A.- Ten (10) outfits (see table VII), unit packaged as specified in 5.1, shall be packed in a fiberboard shipping container, assembled, closed and reinforced conforming to class weather-resistant, size 3A of MIL-B-17757. The fiberboard for the box liner shall conform to type CF, class weather-resistant, variety DW, grade V15c of MIL-B-17757. Level A bundles shall be packed flat, two in length, one in width and five in depth, with the length of the bundle parallel to the end panels of the container.

5.3.2 Level B.- Ten (10) outfits (see table VII), unit packaged as specified in 5.1, shall be packed in a fiberboard shipping container, assembled, closed and reinforced conforming to class domestic, grade 275, size 3A of MIL-B-17757. The fiberboard for the box liner shall conform to type CF, class domestic, variety DW, grade 275 of MIL-B-17757. Level A bundles shall be packed flat, two in length, one in width, and five in depth, with the length of the bundles parallel to the end panels of the container.

5.3.2.1 When specified (see 6.2), the shipping container shall be grade V3c, V3s or V4s fiberboard box fabricated in accordance with MIL-B-17757 and closed in accordance with the appendix of the box specification.

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5.4 Marking.- In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with MIL-STD-129; in addition TYPE II SHELF-LIFE ITEM marking is also required.

Table VI - Items for 5 Man Pack

Item	Specification	FSN	Amount per shipping container
Gloves, Cloth Chemical Protective	MIL-G-1057	8415-782-3246 (Large)	5 pairs
Socks, Chemical Protective	MIL-S-48	8415-782-3223 (Large)	5 pairs
Liner, Shirt, Chemical Protective	MIL-L-43582	8415-900-8242 (Small)	1
Liner, Shirt Chemical Protective	MIL-L-43582	8415-900-8243 (Medium)	3
Liner, Shirt Chemical Protective	MIL-L-43582	8415-900-8244 (Large)	1
Liner, Trousers Chemical Protective	MIL-L-43579	8415-900-8284 (Small)	1
Liner, Trousers Chemical Protective	MIL-L-43579	8415-900-8285 (Medium)	3
Liner, Trousers Chemical Protective	MIL-L-43579	8415-900-8286 (Large)	1
Hood, Chemical Protective	MIL-H-29115	8415-(to be assigned) (One size)	5

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Table VII - Items for 10 Man Pack

Item	Specification	FSN	Amount per shipping container
Glove, Cloth Chemical Protective	MIL-C-1057	8415-782-3246 (Large)	10 pairs
Socks, Chemical Protective	MIL-S-48	8415-782-3223 (Large)	10 pairs
Liner, Shirt Chemical Protective	MIL-L-43582	8415-900-8242 (Small)	2
Liner, Shirt Chemical Protective	MIL-L-43582	8415-900-8243 (Medium)	5
Liner, Shirt Chemical Protective	MIL-L-43582	8415-900-8244 (Large)	3
Liner, Trousers Chemical Protective	MIL-L-43579	8415-900-8284 (Small)	2
Liner, Trousers Chemical Protective	MIL-L-43579	8415-900-8285 (Medium)	5
Liner, Trousers Chemical Protective	MIL-L-43579	8415-900-8286 (Large)	3
Hood, Chemical Protective	MIL-H-29115	8415-(to be assigned) (one size)	10

6. NOTES

6.1 Intended use.- The impregnated outfit (shirt, trouser, hood, socks and gloves) are intended for wear by Navy personnel for protection against chemical agents. The hood was designed to be compatible for wear with the Navy MARK V gas mask.

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6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Whether first article inspection is required (see 3.7 and 4.3).
c. Selection of applicable levels of packaging and packing (see 5.1, 5.2 and 5.3).
d. When weather-resistant grade fiberboard shipping containers are required for levels B packing (see 5.3.2.1).
e. Sizes of the liner outfit (see 3.1).

(1) Time allowed for supplier submission of samples for Government test and evaluation after award of contract.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time required for the Government to notify the supplier whether or not to proceed with production.
(4) The number of the liner outfits to be produced in the first article inspection.

6.3 Batch.- A batch is defined as that quantity of each component article of liner outfits that have been produced by some unit chemical process or subject to some physical conditioning to make the final impregnated articles substantially uniform.

6.4 Process description.- One process for impregnating chemical protective liner outfits is described in Rocky Mountain Arsenal Standard Operating Procedure. No. SMURM-O-P-8, dated 3 December 1970.

Custodian:
Navy - SA

User activities:
Navy - AS, CG, OS, SH, YD

Preparing activity:
Navy - SA

Project No. 8415-N773

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-C-29119; Clothing, Chemical Agent Protective, 5 and 10 Man Outfits; Treatment, Packaging and Packing of

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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POSTAGE AND FEES PAID

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